[Lecture Note] Overview of the Data Mining Process

MGS3701 Data Mining, Spring 2025

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Fri, 23 May 2025

https://www.youtube.com/watch?v=7rs0i-9nOjo

Introduction

- In this chapter, we introduce methods commonly known as "data mining."
- Data mining concentrates mainly on **predictive analytics**, including:
  - *Classification*
  - *Prediction*
  - *Pattern discovery*

Core Idea in Data Mining

Classification

- The process of assigning data into different categories or classes.
  - Examples: Approving/denying loans, identifying fraudulent credit card transactions.
- Classification focuses on cases where the outcome category is unknown or not yet determined.
- Similar techniques include rule-based classification.

Prediction

- Forecasting a numerical outcome or continuous numeric value (e.g., predicting sales amount or future stock prices).
- Instead of categories, it predicts numeric values.

Association Rules and Recommendation Systems

- Identify "what items go together".
- Association rules (also called affinity analysis) uncover common general relationships.
  - Grocery stores: product placement, weekly promotions.
  - Hospitals: predicting symptoms.
  - Recommendation websites: online shopping ("Customers who bought X also bought Y").

> **Collaborative Filtering**
>
> Collaborative filtering uses historical data on users' preferences to recommend items to a specific user.

> **Association Rules vs. Collaborative Filtering**
>
> - **Association rules** find patterns general to a whole group.
> - **Collaborative filtering** generates personalized recommendations by user.

Predictive Analytics

- Predictive analytics includes techniques like classification, prediction, association rules, and collaborative filtering.
- Areas such as clustering can also fit into predictive analytics, depending on context.

Data Reduction and Dimension Reduction

- **Data reduction**: Combines similar records into a smaller number of groups.
  - Example: grouping products or customers.
- **Dimension reduction**: Selects important variables to use, reducing complexity.
  - Helps models work better by simplifying data without losing important information.

Data Exploration and Visualization

- Aims to understand your data, detect unusual patterns, and discover useful relationships.
- Helps with data cleaning, manipulation, visual discovery, and asking good research questions.
- Visual techniques include:
  - **Histograms and boxplots** (numerical data) to learn distributions, identify extremes.
  - **Bar charts** (categorical data) to summarize and visualize groups.
  - **Scatter plots**: detect relationships and outliers between numerical variables.

Supervised and Unsupervised Learning

- A major distinction in data mining methods is between supervised and unsupervised algorithms.

Supervised Learning Algorithms

- Use labeled data (known outcomes).
- Train on data where outcomes (labels) are already known; predict outcomes for new unknown data.
- Common algorithms: Classification models, regression models.

> ℹ Linear Regression as a Supervised Machine Learning Algorithm
>
> - Linear regression models use known values (training data) to predict future unknown outcomes.
> - Example: Predicting house prices given past sales data.

Unsupervised Learning Algorithms

- Do not require labeled data.
- Include **clustering**, **association rules**, and **dimension reduction**.
- Helpful in uncovering structures and patterns even without known outcomes.

> ℹ Note
>
> **Example**: First use clustering to segment loan applicants into risk groups; then apply supervised models to each group to predict default risk.

The Steps in Data Mining

Data Mining Steps

1. **Define project purpose:** Understand business needs, actions intended from results, users of the analysis.
2. **Obtain data:** Sources including internal and external databases.
3. **Explore and clean data:** Handle missing data, ensure units and measurements are consistent, detect obvious errors.
4. **Reduce dimensions if needed:** Remove redundant variables, transform variables or create derived variables.
5. **Choose a data mining task:** Clearly define whether classification, prediction, clustering, etc.
6. **Partition data:** Split data into training, validation, and possibly test sets (for supervised tasks).
7. **Select technique:** Choose algorithms to use (e.g., regression, clustering).
8. **Apply algorithms:** Iterate and refine, trying different settings and versions.

9. **Interpret the results:** Select the best model, confirm performance using validation and test data.
10. **Deploy model:** Implement the chosen model for decision-making or actions.

SEMMA Framework

The above steps resemble SAS's SEMMA methodology, which stands for: - **Sample**, **Explore**, **Modify**, **Model**, **Assess**

> IBM SPSS uses a similar standard procedure called **CRISP-DM** (CRoss-Industry Standard Process for Data Mining).

Other Common Frameworks

- **KDD (Knowledge Discovery in Databases)**: Identifies patterns and relationships from large databases.
- **CRISP-DM**: Consists of Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

https://www.youtube.com/watch?v=q_okDS2RtzY

Preliminary Steps

Organization of Datasets

- Data typically arranged with **variables in columns** and **observations (records) in rows**.
- Example: West Roxbury housing data contain 14 variables (columns) describing over 5000 properties (rows). Outcome variable (home value) is usually first or last variable column.

Predicting Home Values in West Roxbury

- Zillow is a popular real estate valuation site that publishes estimated home ("Zestimate") data.
- Data from city assessments serve as a common basis for home-value prediction models.
- Using publicly available data can inform effective home value predictions.

Loading and Exploring Data in R

- Data loaded into R typically come from CSV files, forming a **data frame** (rows = observations/cases, columns = variables).
- Initial data exploration includes:
  - View data summaries, dimensions.
  - Check variable types and values.

Sampling from Database

- Typically, analyses involve only part of an entire large database—it's faster and often sufficient.
- Random sampling techniques maintain representativeness.
- For rare occurrences, special sampling methods (**oversampling**) can balance datasets better.

Oversampling Rare Events

- Rare events (fraud, purchases) need special sampling strategies.

- Misclassification costs (missing fraud/purchasers) drive special attention:
  - Sampling may intentionally over-represent rare cases.
  - Accept higher false positives if cost of false negatives is high.

Preprocessing and Cleaning the Data

Types of Variables

- Numerical or text variables, continuous, integer, categorical (ordinal/nominal), date/time variables.

Handling Categorical Variables

- Categorical variables often must be converted to numeric dummy (binary) variables for modeling.
- Nominal variables broken down into multiple "dummy" (yes/no) variables.

Variable Selection

- "More is better" isn't always true. Select variables carefully based on domain knowledge.
- Too many variables may lead to complication and poor-performing models (overfitting).

How Many Variables and How Much Data?

- Rule of thumb: number of rows needed depends on predictors used.

- Examples: power calculations, number of variables compared to dataset size.

Outliers and Missing Values

- **Outliers** (extreme values): Identify, verify, and handle.
- **Missing Values**: Decide whether to fill, remove rows or variables, consult experts.

Normalizing (Standardizing) Data

- Data normalized to improve algorithms sensitive to numeric scale.
  - Examples: clustering, nearest-neighbor methods.

Predictive Power and Overfitting

- **Overfitting**: Models become too complex, capturing "noise" instead of true patterns.
- **Generalization**: Model's ability to predict accurately for new data.

Data Partitioning

- **Training set**: Builds the model.
- **Validation set**: Checks accuracy and helps choose best-performing model.
- **Test partition**: Final reliability checking of selected model.

Cross-Validation

- For small datasets, use multiple partitions repeatedly (**cross-validation**) to better estimate performance.

Building a Predictive Model

Has multiple phases including: 1. Define purpose 2. Obtain data 3. Explore and preprocess 4. Reduce dimensions if needed 5. Define data mining task 6. Partition data properly 7. Select and apply algorithms 8. Interpret model results 9. Deploy the chosen predictive model

Using R for Data Mining

- Common for predictive modeling with user-friendly tools to handle preprocessing, analysis, and interpretation phases.

Automating Data Mining Solutions

- Models deployed must run automatically and reliably:
  - Examples: Fraud detection, recommendations systems.
- Deployment requires embedding models in production settings with clear data flows (APIs).
- Model deployment and ongoing maintenance represent real-world challenges beyond just prototyping models.

House management

More to Read

Assignment

- What to do
- Requirement
  - PDF format
  - file name should be include your student id and name
    - stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
- Due date
  - by DATE 11:59PM
  - NO LATE SUBMISSION ALLOWED!!!!

Reference

